COMPUTER ENGINEERING

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

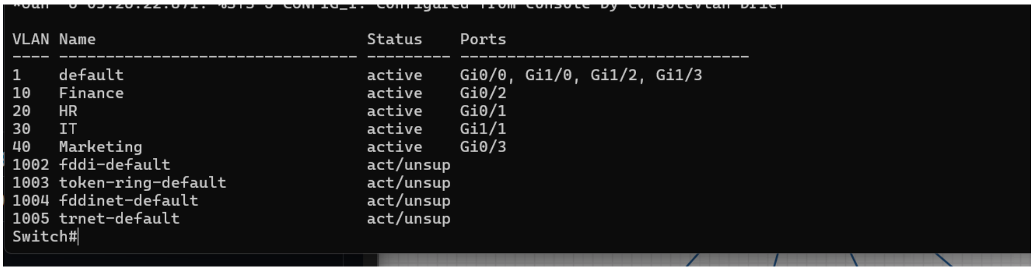
Laporan Akhir
Praktikum Jaringan Komputer

VLAN, Trunk, OSPF, Multivendor

Riggy Fahmi Abyan - 5024241045

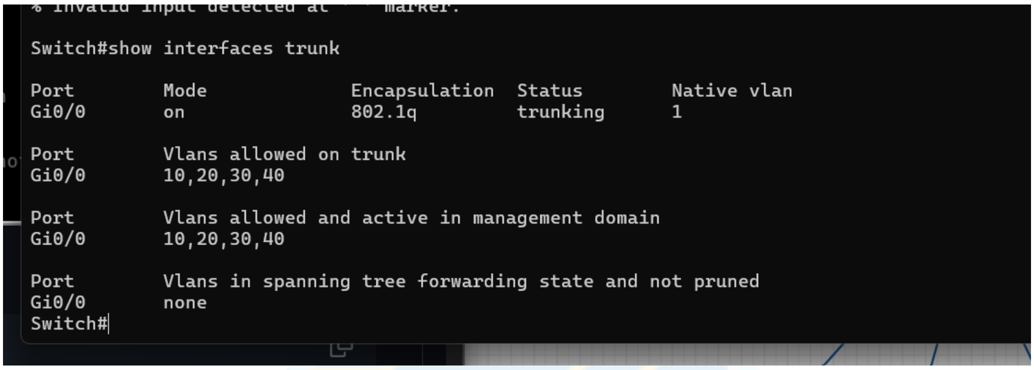
6 Juni 2026

1 Langkah-Langkah Percobaan



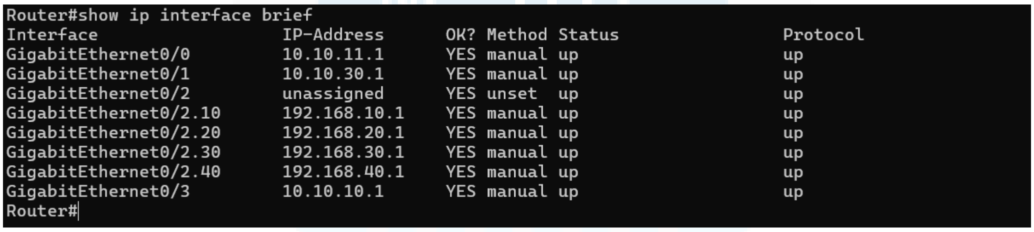
```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/0, Gi1/0, Gi1/2, Gi1/3
10   Finance                          active    Gi0/2
20   HR                               active    Gi0/1
30   IT                               active    Gi1/1
40   Marketing                        active    Gi0/3
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

Gambar 1: membuat VLAN



```
Switch#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/0       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/0       10,20,30,40

Port        Vlans allowed and active in management domain
Gi0/0       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       none
Switch#
```

Gambar 2: Konfigurasi Trunk

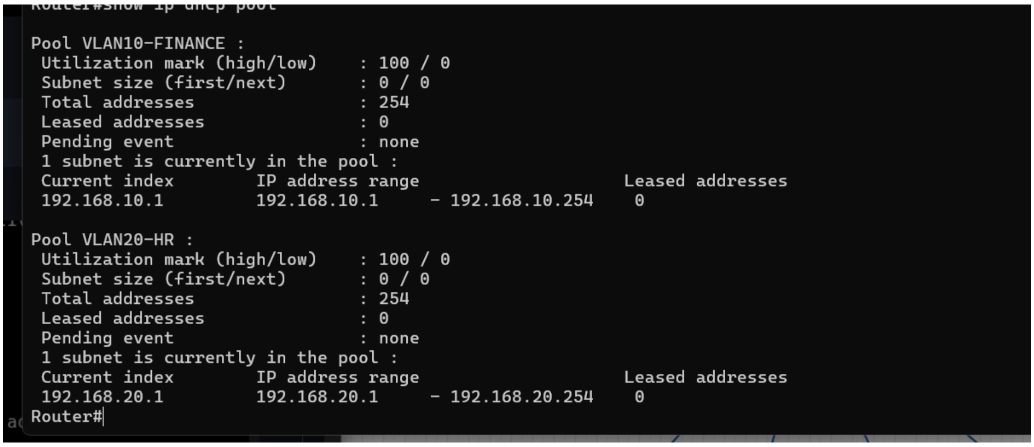


```
Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         10.10.11.1      YES manual up                    up
GigabitEthernet0/1         10.10.30.1      YES manual up                    up
GigabitEthernet0/2         unassigned      YES unset  up                    up
GigabitEthernet0/2.10      192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20      192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30      192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40      192.168.40.1    YES manual up                    up
GigabitEthernet0/3         10.10.10.1      YES manual up                    up
Router#
```

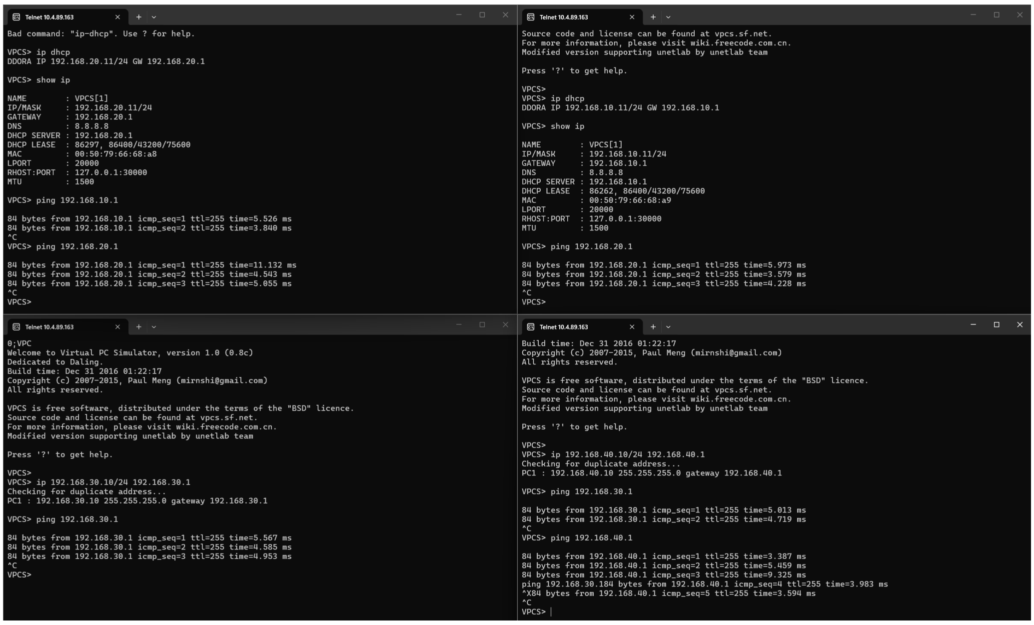
Gambar 3: konfigruasi Cisco ROuter Interface

2 Analisis Hasil Percobaan

Praktikum ini berhasil mengaplikasikan jaringan multivendor dengan isi VLAN, routing OSPF, dan bisa failover menggunakan tiga perangkat berbeda yaitu Cisco, Huawei, dan MikroTik. Hasil pengujian menunjukkan bahwa VLAN 10, 20, 30, dan 40 berhasil dikonfigurasi dengan baik pada Cisco Switch dan Cisco Router sebagai gateway m, di mana setiap VLAN memperoleh IP address sesuai konfigurasi DHCP maupun static yang ditentukan, dan seluruh PC dapat saling terhubung dengan gateway masing-masing. Pada sisi routing antara router, koneksi point-to-point antara Cisco, Huawei, dan MikroTik berhasil dibangun dengan IP address yang sesuai, dan OSPF multivendor berhasil membentuk hubungan neighbor di ketiga perangkat meskipun berasal dari vendor yang berbeda. Pengaturan OSPF cost juga berhasil diterapkan sehingga traffic secara default melewati dua jalur utama Cisco-Huawei, sementara jalur melalui MikroTik hanya digunakan sebagai backup. Pengujian failover pada Praktikum 5 membuktikan bahwa OSPF bekerja secara dinamis: ketika satu link Cisco-Huawei dimatikan, traffic tetap melewati link utama yang masih aktif, dan ketika kedua link utama dimatikan,



Gambar 4: konfigurasi pool cisco router



Gambar 5: pengujian ip VPC dan ping

routing secara otomatis berpindah ke jalur backup melalui MikroTik tanpa perlu konfigurasi manual, kemudian kembali ke jalur utama saat link tersebut diaktifkan kembali.

3 Hasil Tugas Modul

Di Read.me

4 Kesimpulan

Dari paraktikum bisa dipelajari metode yang bisa dipakai untuk backup ajringan. Bisa dilihat bahwa dengan adanya backup, ketika ada link yang dimatikan, jaringan masih bisa bekerja dan tidak putus.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0100.5079.6668.a9       Jun 07 2026 03:34 AM    Automatic
192.168.20.11       0100.5079.6668.a8       Jun 07 2026 03:34 AM    Automatic
Router#
Router#
```

Gambar 6: IP DHCP Binding

```
Telnet 10.4.89.163
Bad command: "ip-dhcp". Use ? for help.

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.20.11/24
GATEWAY     : 192.168.20.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86297, 86400/43200/75600
MAC         : 00:50:79:66:68:a8
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=5.526 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.840 ms
^C
VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=11.132 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=4.543 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=5.055 ms
^C
VPCS>
```

```
Telnet 10.4.89.163

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.10.11/24
GATEWAY     : 192.168.10.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86262, 86400/43200/75600
MAC         : 00:50:79:66:68:a9
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=5.973 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.579 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=4.228 ms
^C
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.218 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=4.482 ms
^C
VPCS>
```

```
Telnet 10.4.89.163
0;VPC
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=5.567 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.585 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=4.953 ms
^C
VPCS>
```

```
Telnet 10.4.89.163
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=5.013 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.719 ms
^C
VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.387 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=5.459 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=9.325 ms
ping 192.168.30.184 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.983 ms
^X84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.594 ms
^C
VPCS>
```

Gambar 7: Show ip dan ping

```
Router(config)#end
Router#conf t
*Jun  6 03:31:47.696: %SYS-5-CONFIG_I: Configured from console by cshow ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/3     unassigned      YES unset  administratively down down
Router#
Router#
```

Gambar 8: Show interface

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; TO-HUAWEI
     10.10.20.2/30      10.10.20.0      ether1
 1   ;;; TO-CISCO
     10.10.30.2/30      10.10.30.0      ether2
```

Gambar 9: ip address print mikrotik

```
*Jun  6 04:01:15.390: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to upping 10.10.
10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/5 ms
Router#
```

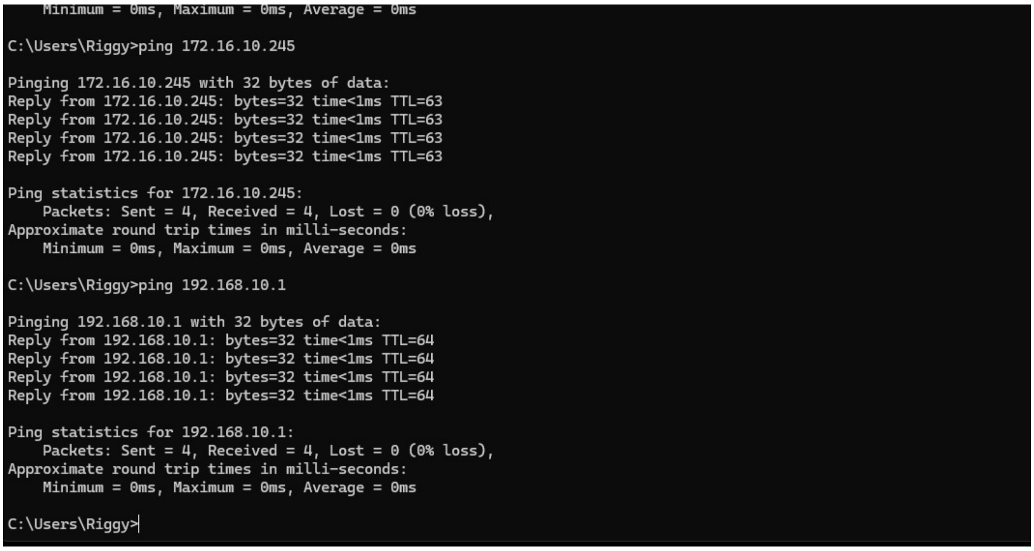
Gambar 10: ping dari cisco router

```
   Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
   Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.10.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 2/2/3 ms

~HUAWEI]
~HUAWEI]ping 10.10.11.1
 PING 10.10.11.1: 56  data bytes, press CTRL_C to break
   Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
   Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

 --- 10.10.11.1 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 2/2/3 ms

~HUAWEI]
~HUAWEI]ping 10.10.20.2
 PING 10.10.20.2: 56  data bytes, press CTRL_C to break
   Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
   Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
   Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
   Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
   Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

 --- 10.10.20.2 ping statistics ---
   5 packet(s) transmitted
   5 packet(s) received
   0.00% packet loss
   round-trip min/avg/max = 1/1/2 ms
```

Gambar 11: ping dari huawei

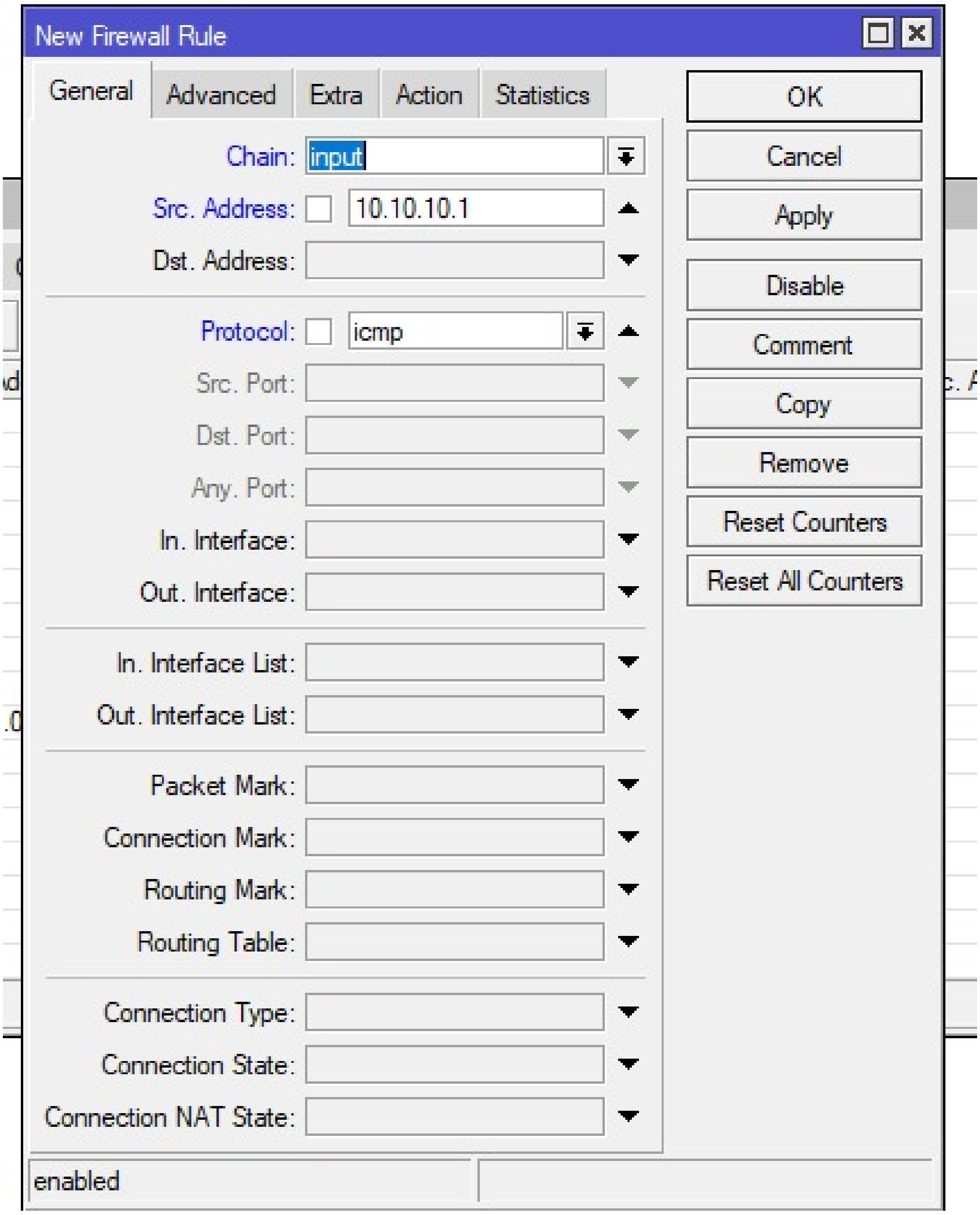
```
         OSPF Process 1 with Router ID 2.2.2.2
               Neighbors

 Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.10.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.10.1        BDR: 10.10.10.2        MTU: 1500
   Dead timer due in  40  sec
   Retrans timer interval: 5
   Neighbor is up for 00h04m45s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]

 Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
 Router ID: 3.3.3.3              Address: 10.10.20.2
   State: Full           Mode:Nbr is Master     Priority: 1
   DR: 10.10.20.1        BDR: 10.10.20.2        MTU: 1500
   Dead timer due in  38  sec
   Retrans timer interval: 5
   Neighbor is up for 00h02m25s
   Neighbor Up Time : 2026-06-06 04:10:04
   Authentication Sequence: [ 0 ]

 Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.11.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.11.1        BDR: 10.10.11.2        MTU: 1500
   Dead timer due in  34  sec
   Retrans timer interval: 5
   Neighbor is up for 00h04m45s
   Neighbor Up Time : 2026-06-06 04:07:43
   Authentication Sequence: [ 0 ]

[~HUAWEI]
```

Gambar 12: konfigurasi OSPF Huawei Router

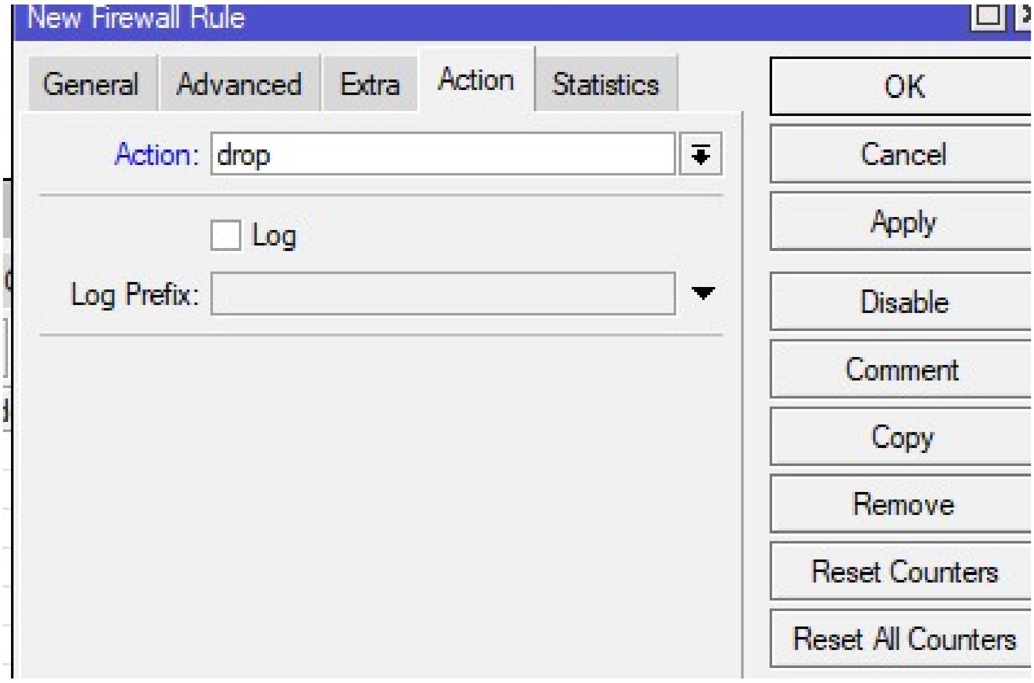


```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riggy>ping 172.16.10.245

Pinging 172.16.10.245 with 32 bytes of data:
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63

Ping statistics for 172.16.10.245:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riggy>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riggy>
```

Gambar 13: ping



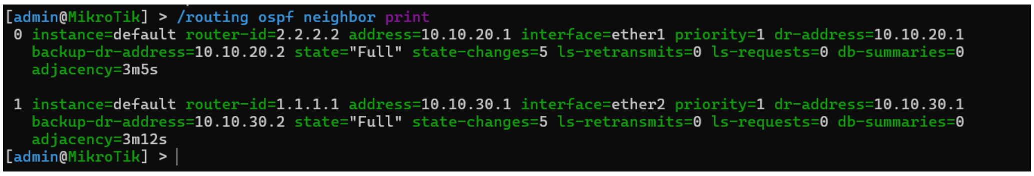
Gambar 14: membuat aturan firewall



Gambar 15: action

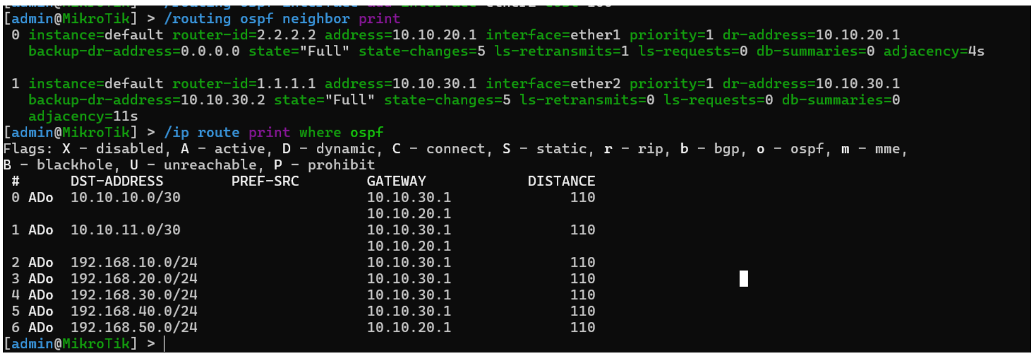
Template Laporan Akhir/P1/h_where.png

Gambar 16: Route OSPF Huawei



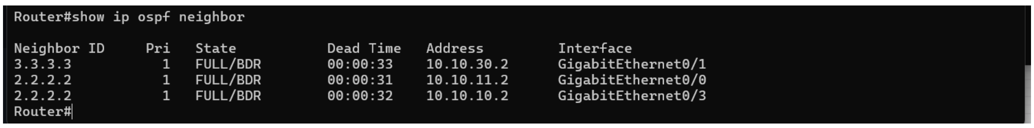
```
[admin@MikroTik] > /routing ospf neighbor print
 0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
   backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m5s

 1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
   backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=3m12s
[admin@MikroTik] > |
```

Gambar 17: Neighbor Mikrotik



```
[admin@MikroTik] > /routing ospf neighbor print
 0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
   backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=1 ls-requests=0 db-summaries=0 adjacency=4s

 1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
   backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
   adjacency=11s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
 #      DST-ADDRESS        PREF-SRC        GATEWAY            DISTANCE
 0 ADo  10.10.10.0/30                      10.10.30.1              110
                                           10.10.20.1
 1 ADo  10.10.11.0/30                      10.10.30.1              110
                                           10.10.20.1
 2 ADo  192.168.10.0/24                    10.10.30.1              110
 3 ADo  192.168.20.0/24                    10.10.30.1              110
 4 ADo  192.168.30.0/24                    10.10.30.1              110
 5 ADo  192.168.40.0/24                    10.10.30.1              110
 6 ADo  192.168.50.0/24                    10.10.20.1              110
[admin@MikroTik] > |
```

Gambar 18: Rute OSPF Mikrotik

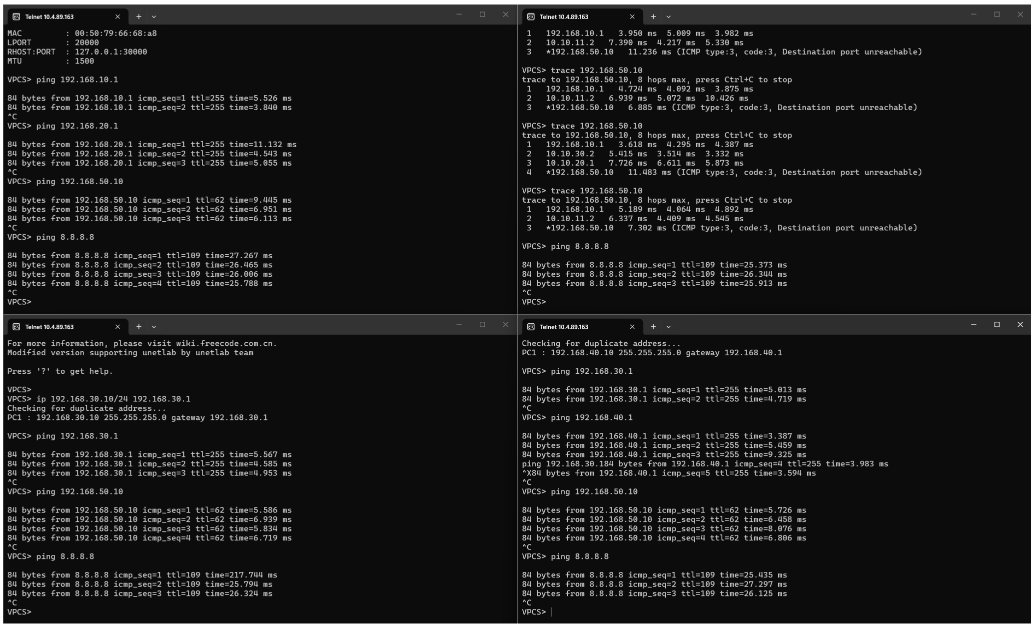


```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:33    10.10.30.2      GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:31    10.10.11.2      GigabitEthernet0/0
2.2.2.2           1   FULL/BDR        00:00:32    10.10.10.2      GigabitEthernet0/3
Router#
```

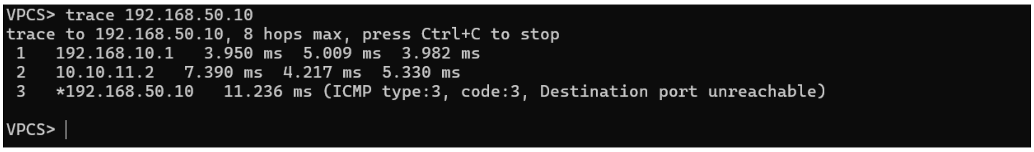
Gambar 19: Neighbor Cisco Router

Template Laporan Akhir/P1/h_ospf.png

Gambar 20: Neighbor sudah ada

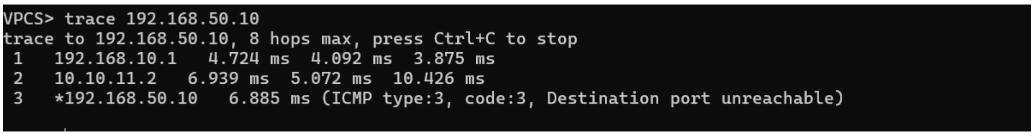


Gambar 21: ping dari VPC



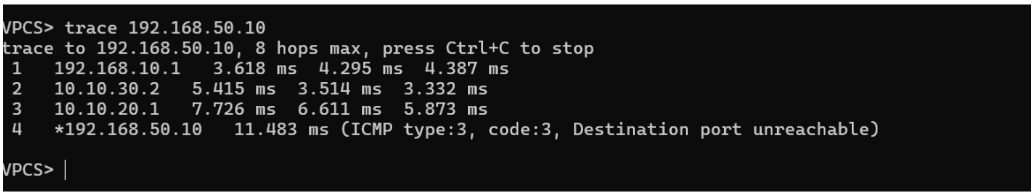
```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   3.950 ms  5.009 ms  3.982 ms
 2   10.10.11.2   7.390 ms  4.217 ms  5.330 ms
 3   *192.168.50.10   11.236 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 22: Trace dari VPC menuju Huawei



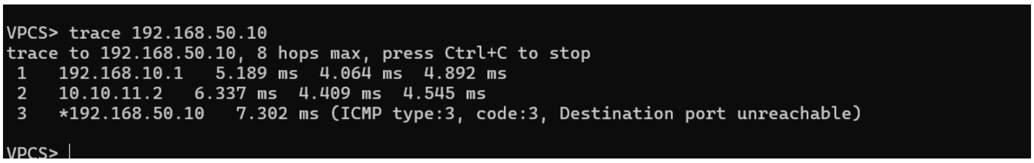
```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   4.724 ms  4.092 ms  3.875 ms
 2   10.10.11.2   6.939 ms  5.072 ms  10.426 ms
 3   *192.168.50.10   6.885 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 23: Trace dari VPC ke Huawei dengan kondisi berbeda



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   3.618 ms  4.295 ms  4.387 ms
 2   10.10.30.2   5.415 ms  3.514 ms  3.332 ms
 3   10.10.20.1   7.726 ms  6.611 ms  5.873 ms
 4   *192.168.50.10   11.483 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 24: Trace dari VPC ke Huawei dengan kondisi berbeda



```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1   192.168.10.1   5.189 ms  4.064 ms  4.892 ms
 2   10.10.11.2   6.337 ms  4.409 ms  4.545 ms
 3   *192.168.50.10   7.302 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 25: Trace dari VPC ke Huawei dengan kondisi berbeda